

GBOLAGADE KAZEEM ADEWALE

Physiology Researcher | Neuroscience Education | Serious Games & Full-Stack Development

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ACADEMIC PROFILE

Physiology and neuroscience researcher with peer-reviewed publications in physiology education and experience independently programming and deploying research-informed educational games. Combines training in human physiology, neurophysiology and quantitative analysis with full-stack development using JavaScript and Firebase. Current interests include technology-enhanced physiology education, serious games, neurophysiology, pain and neuroinflammation, and dopamine-mediated motivation and reward.

RESEARCH INTERESTS

Physiology education and assessment; serious games and gamification; technology-enhanced medical education; neurophysiology; pain and neuroinflammation; dopamine, motivation and reward; learning analytics; low-bandwidth digital learning for resource-limited settings.

EDUCATION

- M.Sc. Physiology (in progress)** | University of Ilorin, Ilorin, Nigeria 2025-Present
- Research area: Neurophysiology.
- B.Sc. Physiology, Second Class Honours (Upper Division)** | University of Medical Sciences, Ondo, Nigeria 2017-2021
- Undergraduate thesis: Effects of nonanoic acid on cadmium-induced reproductive toxicity in male Wistar rats.
 - Graduated with distinctions in Neurophysiology and Neuroanatomy.

PEER-REVIEWED PUBLICATIONS

Alagbonsi, A. I., Salman, T. M., Dada, M. O., Okedun, E. S., Ajibola, A. A., Adepoju, A. A., **Gbolagade, K. A.**, & Marcondes, F. K. (2026). GutDigest: A new game to enhance active learning and assessment of gastrointestinal digestion of macromolecules. *Advances in Physiology Education*, 50(3), 804-812. doi:10.1152/advan.00308.2025

Alagbonsi, A. I., Salman, T. M., Assayouti, S. B., Ahmed, A., Aribatise, B., **Gbolagade, K. A.**, Sulaiman, S. O., & Marcondes, F. K. (2026). GutMove: A new educational game for active learning and assessment of competencies on gastrointestinal motility. *Advances in Physiology Education*, 50(3). doi:10.1152/advan.00007.2026

RESEARCH SOFTWARE AND EDUCATIONAL GAMES

- GutDigest - Gastrointestinal Digestion of Macromolecules** | Research software 2026
- [Live game](#) | Published in *Advances in Physiology Education*
- Independently programmed and deployed the web application within a collaborative physiology-education project.
 - Built an interactive matching environment covering carbohydrate, lipid and protein digestion, immediate feedback, reshuffled trials and mobile-friendly interaction.
- GutMove - Gastrointestinal Motility Physiology Gamification** | Research software 2026
- [Live game](#) | Published in *Advances in Physiology Education*
- Independently programmed and deployed the application within a collaborative physiology-education project.
 - Implemented seven stages and 75 tasks, authentication, scorecards, result verification, mistake review, responsive controls and accessibility-oriented display options.
- GutAcid - Gastric Acid Physiology Game** | Research software 2026
- [Live game](#)
- Designed, programmed and deployed an interactive learning game focused on gastric acid secretion, regulation and related gastrointestinal physiology.
- All or None - Neurophysiology Reasoning Game** | Independent project 2026
- [Live game](#)
- Designed, programmed and deployed a confidence-based neurophysiology game built around conceptual discrimination, retrieval practice and immediate performance feedback.

RESEARCH AND ACADEMIC EXPERIENCE

- EdTech Researcher and Lead Technical Developer** | Department of Physiology, University of Ilorin 2025-Present
- Collaborate with physiology educators to translate difficult biomedical concepts into interactive learning and assessment tools.
 - Contributed to the design, implementation and scholarly reporting of GutDigest and GutMove, resulting in two peer-reviewed publications.
 - Develop and test web-based educational systems intended for medical and health-science students, including mobile-first and resource-conscious delivery.

Independent Neuroscience Researcher | Neuropsychology of motivation and addiction

2025-Present

- Develop conceptual and review-based work on dopamine, craving, motivational design and the redirection of compulsive drive toward constructive outcomes.
- Conduct literature synthesis and theory development across neuroscience, psychology, reward learning and behavioural addiction.

PREPRINTS AND SCHOLARLY WORKING PAPERS

Gbolagade, K. A. (2025). Harnessing addiction: A neuropsychological framework for rechanneling compulsive drive toward positive outcomes [Preprint]. *PsyArXiv*. doi:10.31234/osf.io/mpczn_v1

Gbolagade, K. A. (2025). Rechanneling craving: A neuropsychological framework for constructive addiction and motivational design [Preprint]. *PsyArXiv*. doi:10.31234/osf.io/ytgv4_v1

PROFESSIONAL AND TECHNICAL EXPERIENCE

Lead Software Architect and Full-Stack Developer | Independent Consultant

2024-Present

- Design and deploy custom web applications, administrative systems and secure content-management workflows for institutional and professional clients.
- Built a bespoke platform for Juriscribe Partners with a custom content-management system, Firebase integration and client-controlled publishing workflow.
- Develop responsive interfaces, authentication flows, database structures and operational dashboards without reliance on generic templates.

Academic Director | Private Educational Institution, Iwo, Osun State

2024-Present

- Oversee curriculum delivery, academic operations and student-performance monitoring.
- Designed and deployed a computer-based testing system that randomises JAMB questions and automates grading and eligibility decisions using class-specific cut-offs.

SELECTED CLINICAL EXPERIENCE

Vital Sign Specialist / Clinical Physiologist (NYSC) | The Polytechnic Ibadan School Health Centre

2022-2023

- Performed physiological assessments, recorded patient vital signs before clinical consultation and supported health-data documentation.

Clinical Physiologist Intern | General Hospital Ondo

2019-2020

- Supported diagnostic and therapeutic workflows involving cardiovascular and neurological presentations in a high-volume clinical environment.

TECHNICAL AND RESEARCH SKILLS

Research: Literature reviews; scientific writing; quantitative data analysis; study conceptualisation; educational-tool evaluation; clinical data documentation.

Data analysis: Python; R; SPSS.

Software development: HTML; CSS; JavaScript; responsive UI/UX; Firebase Authentication; Cloud Firestore; Firebase Hosting; custom CMS and administrative dashboards.

Physiology and neuroscience: Human physiology; neurophysiology; gastrointestinal physiology; pain and neuroinflammation; motivation and reward systems.

Clinical: Vital-sign assessment; patient triage support; medical-record management; clinical clearance workflows.